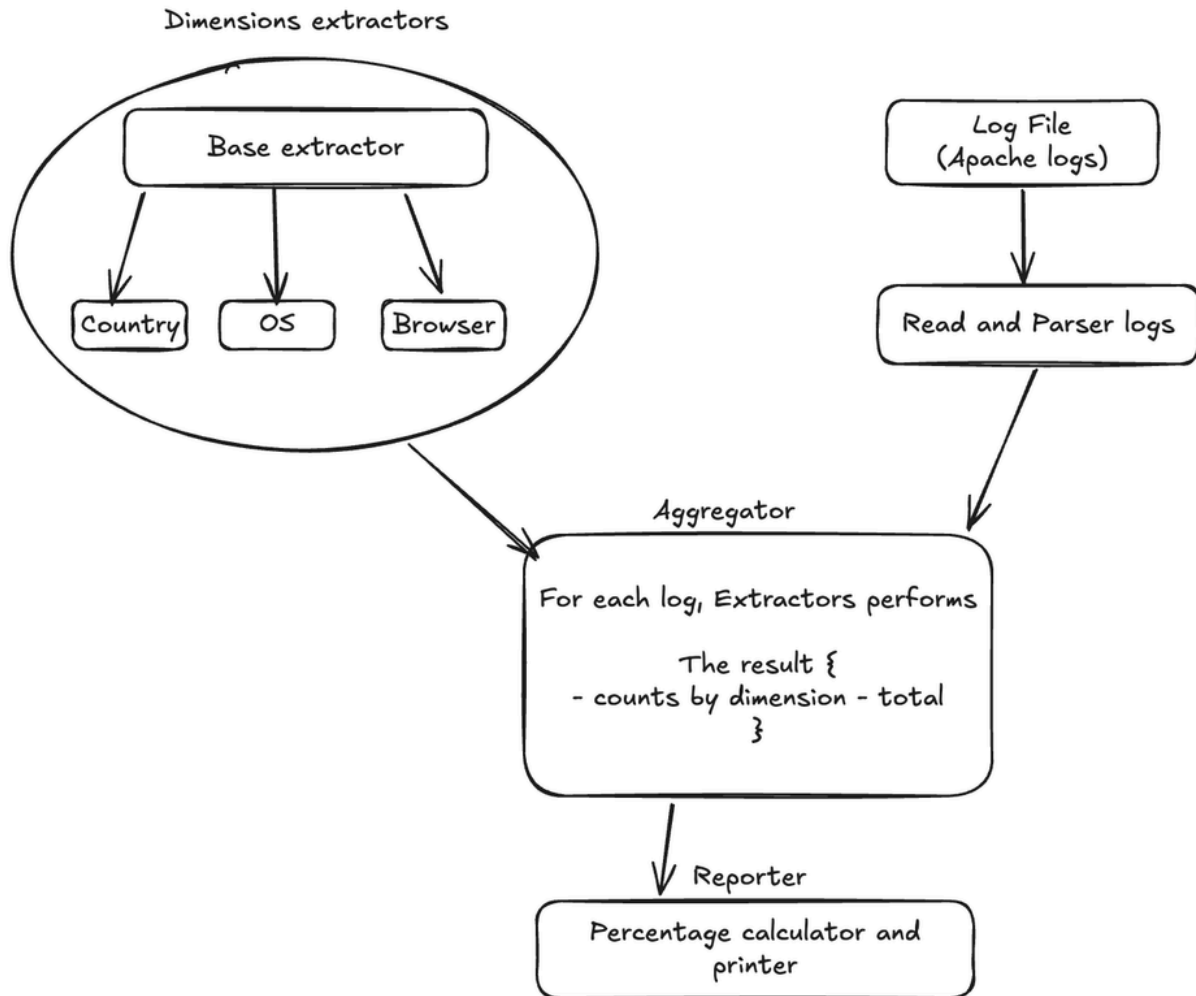


Statistical Reporting Module – Design Document



Module Flow

1. The Apache Log file is read line by line by the **Parser**, which extracts the IP and User-Agent from each line. The **GeoIP2 database** is accessed **once** at the start, and closed at the end of processing.
2. Each line is passed to the different **Extractors** (Country, OS, Browser), which return the corresponding values or "Unknown" if the value cannot be identified.
3. The **Aggregator** updates the counts for each dimension and keeps track of the total number of lines (total).
4. After processing, the **Report Generator** function calculates percentages for each value and prints a well organized report sorted by frequency.

Extensibility

- **Adding a new dimension:** Simply create a new Extractor that inherits from **DimensionExtractor** and add it to the **extractors** list of the **Aggregator** it will automatically be included in processing and in the report.
- **Adding a new report type:** Create a new function that receives **aggregator.counts** (or total) and processes the data into the desired format, such as JSON, CSV, or HTML – it can be used alongside the existing report.

